2D Two-Phase Reservoir Simulator: Physics, Validation, and Asset Economics

A finite-volume, IMPES (Implicit Pressure, Explicit Saturation) simulator for incompressible oil-water flow in a horizontal reservoir, validated against the Buckley-Leverett analytical solution and coupled to a discounted cash flow model that optimizes flood design on NPV. This document states the physics, the discretization, and the numerical choices, with the reasoning behind each one, then presents the validation, the results, and the economics.

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1. Problem statement

The model simulates a waterflood: water is injected into an oil-filled reservoir to push oil toward a producing well. The simulator tracks two fields on a 2D grid through time, the pressure p and the water saturation S_w, and reports how much oil is recovered and when water breaks through at the producer.

The test case is a quarter five-spot: an injector at one corner and a producer at the opposite corner, with no-flow outer boundaries.

2. Assumptions

The governing equations below follow from a deliberate set of simplifications. Each is standard for a first-principles waterflood model and each is defensible.

- Two immiscible phases, water and oil, no mass transfer between them.
- Incompressible rock and fluids (constant porosity, constant densities).
- Horizontal, areal (2D) flow, so gravity is neglected.
- Zero capillary pressure, so both phases share a single pressure field p.
- Isothermal.
- Absolute permeability may be homogeneous or spatially variable; porosity is constant.

The zero-capillary and no-gravity assumptions are the two most significant. They keep the pressure equation to a single unknown pressure per cell and are appropriate for a teaching-scale areal waterflood. Both can be added later without changing the overall structure.

3. Governing equations

3.1 Darcy's law (per phase)

Each phase moves down the pressure gradient at a rate set by absolute permeability k, that phase's relative permeability $k_{r\alpha}$, and its viscosity μ_α:

$$\mathbf{u}_\alpha = -\frac{k\,k_{r\alpha}}{\mu_\alpha}\,\nabla p = -\lambda_\alpha\,k\,\nabla p, \qquad \alpha = w, o$$

where $\lambda_\alpha = k_{r\alpha}/\mu_\alpha$ is the phase mobility.

3.2 Mass conservation (per phase)

For an incompressible phase, accumulation in the pore space balances the net flux out plus any well source or sink:

$$\phi \frac{\partial S_\alpha}{\partial t} + \nabla \cdot \mathbf{u}_\alpha = q_\alpha$$

with the saturation constraint $S_w + S_o = 1$.

3.3 Splitting into a pressure equation and a saturation equation

Adding the water and oil conservation equations and using $S_w + S_o = 1$ (so the time-derivative terms cancel for incompressible flow) removes saturation from the sum and yields the **pressure equation**:

$$-\nabla \cdot (\lambda_t \, k \, \nabla p) = q_t, \qquad \lambda_t = \lambda_w + \lambda_o$$

This is elliptic. It has no time derivative, which is the mathematical statement that in an incompressible medium a pressure change is felt everywhere at once. This is why pressure is solved implicitly across the whole grid.

The water equation, rewritten with the total velocity $\mathbf{u}_t = -\lambda_t k \nabla p$ and the fractional flow f_w, is the **saturation (transport) equation**:

$$\phi \frac{\partial S_w}{\partial t} + \nabla \cdot (f_w \, \mathbf{u}_t) = q_w, \qquad f_w = \frac{\lambda_w}{\lambda_w + \lambda_o}$$

This is hyperbolic. It describes a saturation front advancing at a finite speed, which is why saturation can be updated explicitly and locally.

The split of one elliptic pressure equation and one hyperbolic saturation equation is the entire basis of the IMPES method.

3.4 Relative permeability (Corey model)

Relative permeabilities depend on a normalized water saturation that maps the mobile range $[S_{wc}, 1 - S_{or}]$ onto $[0, 1]$:

$$S_{wn} = \frac{S_w - S_{wc}}{1 - S_{wc} - S_{or}}$$

$$k_{rw} = k_{rw}^{\max} \, S_{wn}^{n_w}, \qquad k_{ro} = k_{ro}^{\max} \, (1 - S_{wn})^{n_o}$$

As water saturation rises, k_{rw} increases (water flows more easily) and k_{ro} decreases (oil is choked off).

3.5 Mobility ratio

The end-point mobility ratio characterizes the displacement:

$$M = \frac{\lambda_w^{\max}}{\lambda_o^{\max}} = \frac{k_{rw}^{\max}/\mu_w}{k_{ro}^{\max}/\mu_o}$$

$M > 1$ is an unfavorable displacement: the injected water is more mobile than the oil it displaces, so it fingers ahead, breaks through early, and leaves oil behind (poor sweep efficiency). $M < 1$ is favorable and sweeps more like a piston.

4. Discretization

A cell-centered finite-volume scheme on a uniform $n_x \times n_y$ grid. Each cell has bulk volume $V = \Delta x\, \Delta y\, h$.

4.1 Transmissibility

The transmissibility of a face measures how easily fluid crosses between two adjacent cells. The geometric (rock-and-geometry) part in the x-direction is:

$$T^{\text{geo}}_{i+\frac{1}{2},j} = \beta\, k_{i+\frac{1}{2},j}\, \frac{\Delta y\, h}{\Delta x}$$

with the field-units constant $\beta = 1.127 \times 10^{-3}$. The face permeability uses the **harmonic mean** of the two neighboring cell permeabilities:

$$k_{i+\frac{1}{2},j} = \frac{2\, k_{i,j}\, k_{i+1,j}}{k_{i,j} + k_{i+1,j}}$$

The harmonic mean is used because flow across the face passes through both cells in series, and series flow is limited by the tighter (lower-permeability) rock. An arithmetic mean would overstate the effective permeability. The geometric transmissibility depends only on rock and geometry, so it is computed once.

The full transmissibility used in the flow equations multiplies in the total mobility at the face, evaluated as the arithmetic average of the two cells:

$$T_{i+\frac{1}{2},j} = T^{\text{geo}}_{i+\frac{1}{2},j}\, \lambda_t^{\text{face}}$$

Mobility depends on saturation, which changes every step, so this part is recomputed each step. (Arithmetic averaging of the face mobility for the pressure equation is a common simplification; upstream weighting is the more rigorous alternative.)

4.2 Pressure equation: implicit solve

Applying mass conservation to each cell (net flux to neighbors equals the well source) gives one linear equation per cell:

$$\sum_{\text{faces}} T_f\,(p_{\text{nb}} - p_c) + q_c = 0$$

Assembled over all $N = n_x n_y$ cells this is the linear system

$$\mathbf{A}\,\mathbf{p} = \mathbf{b}$$

The matrix **A** has each cell's neighbor transmissibilities as positive off-diagonal entries and the negative sum of those transmissibilities on the diagonal. The system is solved directly (sparse solver) so every cell pressure is obtained simultaneously, which is the implicit step.

4.3 Well and boundary conditions

- **Injector:** rate-controlled. A fixed water rate Q_{inj} enters the right-hand side **b** at the injector cell.
- **Producer:** pressure-controlled (Dirichlet). The producer cell's equation is replaced by $p = p_{\text{ref}}$.
- **Outer boundaries:** no-flow (no transmissibility across the domain edge).

Fixing the producer pressure is not optional. With both wells on rate control the system would define only pressure differences, not absolute levels, leaving the solution non-unique (a singular matrix). One fixed pressure anchors the field and makes the solution unique.

4.4 Saturation equation: explicit update with upstream weighting

With pressure known, the signed total flux across each face follows from Darcy's law:

$$q_f = T_f\,(p_c - p_{\text{nb}})$$

The water flux across a face uses the fractional flow evaluated at the **upstream** cell, the cell the flow is coming from:

$$q_{w,f} = f_w(S_{\text{up}})\,q_f$$

Upstream weighting is used because the fluid crossing a face physically carries the upstream cell's composition. Using a downstream or averaged value numerically smears the saturation front; upstream weighting keeps it sharp.

Summing the net water flux over each cell's faces (plus well terms) and stepping forward in time gives the explicit update:

$$S_w^{n+1} = S_w^n + \frac{\Delta t \cdot 5.615}{\phi V}\ \text{(net water rate into cell, bbl/day)}$$

where 5.615 converts barrels to cubic feet.

4.5 Stability: the CFL condition

Because the update is explicit and local, information propagates only one cell per time step. The saturation front must therefore not travel more than about one cell per step, which sets a maximum stable time step:

$$\Delta t \leq \text{CFL} \cdot \frac{\phi V}{5.615 \cdot q_{\max} \cdot \max \left| \frac{df_w}{dS_w} \right|}$$

The CFL safety factor (0.5 here) keeps the step comfortably below the limit. Exceeding this limit makes the front outrun the scheme's one-cell reach and the solution goes unstable (nonphysical oscillations, saturations outside physical bounds).

5. Reported quantities

- **Pore volumes injected:** $\text{PVI} = \frac{V_{\text{inj,cum}} \cdot 5.615}{V_{p,\text{total}}}$, the natural time axis for a waterflood.
- **Water cut** at the producer: f_w evaluated at the producer cell, the fraction of produced fluid that is water.
- **Movable oil in place:** $\text{OOIP} = \frac{(1 - S_{wc} - S_{or})\, V_{p,\text{total}}}{5.615}$ (bbl).
- **Recovery factor:** cumulative produced oil divided by movable OOIP.

6. Solution algorithm (IMPES loop)

1. Initialize $S_w = S_{wc}$ everywhere (reservoir full of oil).
2. Solve the pressure equation implicitly at the current saturation.
3. Compute face fluxes from the pressure field.
4. Choose a stable time step from the CFL condition.
5. Update water saturation explicitly with upstream fractional flow.
6. Record water cut, recovery, and snapshots.
7. Repeat from step 2 until the target pore volumes have been injected.

7. Validation against the Buckley-Leverett analytical solution

A simulator is only worth trusting if it can reproduce a known answer. The 1D Buckley-Leverett problem (incompressible, immiscible, zero capillary pressure) has an exact analytical solution, so the simulator is run on a 1D grid with the base-case fluid properties and compared against it.

The analytical solution comes from the Welge tangent construction. The shock (front) saturation S_{wf} is where the tangent from $(S_{wc}, 0)$ touches the fractional flow curve, satisfying $f'_w(S_{wf}) = f_w(S_{wf})/(S_{wf} - S_{wc})$. Behind the front is a rarefaction: each saturation $S \geq S_{wf}$ travels at its own characteristic speed, $x_D = t_D\, f'_w(S)$, with t_D measured in pore volumes injected. Breakthrough occurs at $\text{PVI}_{bt} = 1/f'_w(S_{wf})$, and the recovery at breakthrough follows from the Welge average saturation $\bar{S}_w = S_{wf} + (1 - f_w(S_{wf}))/f'_w(S_{wf})$.

Results for the base fluids ($M = 1.56$, 200 cells):

Quantity	Analytical	Numerical	Error
Shock saturation S_{wf}	0.575	matches profile	-
Breakthrough (PVI)	0.462	0.456	-1.2%
Recovery factor at breakthrough	0.770	0.760	-1.2%

The numerical saturation profiles sit on top of the analytical rarefaction and shock at both 0.15 and 0.30 PVI. The only visible difference is smearing of the shock over a few cells, the expected numerical diffusion of first-order upwind transport. A grid refinement study (50 to 400 cells) shows the front sharpening toward the analytical discontinuity as Δx shrinks, confirming the scheme converges to the correct solution.

Every run also reports a discrete water material balance,

$$\varepsilon_{MB} = \frac{W_{\text{inj}} - W_{\text{prod}} - \Delta W_{\text{stored}}}{W_{\text{inj}}}$$

which comes back at machine precision ($\sim 10^{-14}$) in all cases. Mass is conserved by construction, not by luck.

8. Results

8.1 Base case

The homogeneous quarter five-spot ($M = 1.56$) breaks through at 0.37 PVI and recovers 84% of movable oil by 1 PVI. The saturation snapshots show the characteristic five-spot front: radial near the injector, then cusping toward the producer corner as the pressure field focuses the flow.

8.2 Mobility ratio

Sweeping oil viscosity from 0.5 to 10 cp spans end-point mobility ratios from 0.19 (favorable) to 3.89 (strongly unfavorable):

μ_o (cp)	M	Breakthrough (PVI)	RF at 1 PVI
0.5	0.19	0.56	0.97
2.0	0.78	0.44	0.90
5.0	1.94	0.35	0.81
10.0	3.89	0.28	0.74

Rising M pulls breakthrough earlier and cuts ultimate recovery, exactly the piston-to-fingering transition the fractional flow theory predicts.

8.3 Heterogeneity

Three permeability fields with the same 200 mD average, same fluids ($M = 1.56$):

Field	Breakthrough (PVI)	RF at 1 PVI
Homogeneous	0.37	0.84
High-perm channel (injector to producer)	0.10	0.51
Correlated log-normal ($V_{DP} = 0.7$)	0.25	0.79

The channel case is the striking one: with identical average permeability, water short-circuits down the channel, breaks through at one tenth of a pore volume, and strands the off-channel oil, cutting recovery by a third. The lesson is that the average permeability of an asset says almost nothing about its flood performance; contrast and connectivity control sweep. This is why reserves attributed to a waterflood depend on geology, not just volumetrics.

9. Layer 2: asset economics and NPV optimization

The production stream from the simulator feeds a monthly discounted cash flow model: flat price deck, net revenue interest, fixed and variable opex, water handling cost on both injected and produced water, and capex split into drilling plus rate-dependent facilities. Production truncates at the economic limit, the first month operating cash flow goes negative, so estimated ultimate recovery (EUR) is an output of the economics rather than the physics alone.

A useful property of incompressible flow makes rate optimization essentially free: the saturation solution depends only on pore volumes injected, so one simulation provides the water cut and recovery curves for every injection rate, each with its own time axis $t = \mathrm{PVI} \cdot V_p/(5.615\,Q)$.

Sweeping the injection rate exposes the central trade-off. Higher rates pull the same barrels forward in time (less discounting) but require larger facilities capex, while low rates burn fixed opex for years. With the base assumptions (70 $/bbl, 25% royalty, 10% discount rate) NPV peaks at an interior optimum of 750 bbl/day, while EUR keeps creeping upward all the way to the highest rate tested. Designing the flood for maximum barrels instead of maximum value would surrender about $2.2MM of NPV on a $5.6MM asset. More oil is not more value, and the gap between the two is where acquisition and divestiture judgment lives.

10. Limitations and extensions

The model is deliberately minimal, and the honest list of what it leaves out is part of understanding it:

- **No compressibility.** Real reservoirs store energy in fluid and rock compression; this model has no primary depletion, only displacement. Adding compressibility would also break the rate-scaling shortcut used in the economics layer.
- **No gravity or capillary pressure.** Fine for a thin areal flood; wrong for thick reservoirs or transition zones. Capillary pressure would also regularize the saturation shock physically rather than numerically.
- **Simple well model.** The producer is a Dirichlet cell rather than a Peaceman well index, so near-well pressure is grid-dependent. A Peaceman model is the standard fix.
- **First-order transport.** Upwind differencing smears the front over a few cells. Higher-order TVD schemes would sharpen it at the same grid size.
- **IMPES stability.** The explicit saturation update carries a CFL limit; a fully implicit or adaptive-implicit formulation would allow larger steps.

- **Flat price deck and deterministic inputs.** The natural next layer is Monte Carlo over permeability, OOIP, and price to produce P10/P50/P90 distributions of reserves and value, which is how assets are actually evaluated in A&D.

11. Nomenclature

Symbol	Meaning	Units
p	pressure	psi
S_w, S_o	water, oil saturation	fraction
S_{wc}, S_{or}	connate water, residual oil	fraction
S_{wn}	normalized water saturation	fraction
k	absolute permeability	mD
$k_{r\alpha}$	relative permeability of phase α	fraction
μ_α	viscosity of phase α	cp
λ_α	phase mobility, $k_{r\alpha}/\mu_\alpha$	1/cp
λ_t	total mobility	1/cp
f_w	fractional flow of water	fraction
M	end-point mobility ratio	dimensionless
ϕ	porosity	fraction
T_f	face transmissibility	bbl/day/psi
q	volumetric rate (well or face)	bbl/day
β	Darcy constant (field units)	1.127e-3
V	cell bulk volume	ft^3
V_p	pore volume	ft^3

Implemented in Python (NumPy, SciPy sparse solver, Matplotlib, pandas). The pressure solve uses a sparse direct solver; the transport update is fully vectorized. Code, studies, and figures: see the repository README.